

Daniel Vargas Díaz

Statement of Objectives, The joint EPFL – ETH Zurich Doctoral Program in the Learning Sciences (JDPLS)

My first experience as an instructor was during my high school years. My school hosted an Intel Club House site, a project aimed at providing STEAM skills to underprivileged populations in various countries. At that time, they acquired a LEGO robotics kit but lacked someone experienced in using and teaching it. Coming from an underprivileged background myself, this was my first encounter with such a kit. I viewed this as an opportunity to not only fulfill my childhood dream of playing with these kits but also to become an instructor in my community, helping other children and teenagers learn about robotics. This experience paved the way for me to teach others in different settings and environments, allowing me to witness firsthand the transformative power of education, as it had on me, a first-generation student whose parents worked tirelessly to provide valuable education despite their limited formal education.

Throughout my academic journey, from undergraduate studies in Electrical Engineering at the University of Costa Rica to my master's degree in Computer Science with a focus on Human-Computer Interaction at Virginia Tech, education has been a guiding force. During my undergraduate years, I collaborated with professors in Special Education to prototype low-cost tools for students with disabilities. This work included building a light table for students with visual impairments and modifying toys with Arduinos to make them interactive with sounds and lights, enhancing the children's learning process. Additionally, I had the privilege of working with the MIT Media Lab for nearly two years, where I gained valuable insights into Human-Computer Interaction and its potential impact on people's lives.

This experience inspired me to pursue a master's degree at Virginia Tech. Here, I have been involved in research projects with the Department of Human Development, focusing on using technology to enhance learning. My thesis, 'TaleMate,' is an example to this endeavor. It is a platform that reimagines the shared reading experience for parents and children, aiming to make reading electronic books more appealing and to promote parent-child interaction, which is currently rare in electronic book reading. Additionally, I developed a Voice Agent for math learning, a project initiated by another graduate student in the Human Development department. My research also extends to understanding the impact of ChatGPT on Computer Science education, involving evaluations of ChatGPT's accuracy and strategies for adapting course materials and engaging with instructors.

As I aim to join the Joint EPFL – ETH Zurich Doctoral Program in the Learning Sciences (JDPLS), I see it as an opportunity to further my mission. The interdisciplinary nature of JDPLS aligns perfectly with my goals, providing a platform to develop innovative solutions for learning challenges in the digital age. During my Ph.D., I plan to combine my electrical and computer engineering background with my HCI and computer science experience to create new technologies for enhancing learning. I am particularly interested in designing embodied and tangible interfaces for students, applying digital fabrication techniques to develop accessible technology. One project idea is to create adaptive smart desks based on the Montessori concept, designed to grow with students and cater to their evolving needs while supporting their learning. These desks could be specialized, such as for students interested in electronics, featuring components that allow for hands-on learning about electronics, direct and alternating current, motors, and smart homes. However, I am also open to exploring other projects that merge my hardware and software expertise to develop new educational tools.

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Upon completing the JDPLS program, my ultimate goal is to become a pivotal figure in education in my home country, revolutionizing learning experiences by integrating advanced technology with education. I envision starting a company that offers consulting services to governmental institutions and schools, enhancing educational quality for children. The JDPLS program is a crucial step in equipping me with the necessary skills and knowledge to effect meaningful change in educational technology and realize my aspirations.